A MAP OF SPECTRAL SPACES IS SPECTRAL EXACTLY WHEN IT IS CONTINUOUS FOR THE CONSTRUCTIBLE TOPOLOGY

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The statement that a spectral map is continuous in the constructible topology and has quasi-compact fibres. Invoked at the start of the counted proof (source line 4628). Gives the easy direction.
- The statement that the constructible topology of a spectral space is Hausdorff and quasi-compact. Invoked inside the counted proof (source lines 4633-4635). Makes the preimage of a quasi-compact open quasi-compact in the constructible topology.
- The statement that quasi-compactness transfers across a surjective continuous map. Invoked at the end of the counted proof (source line 4640). Transfers quasi-compactness back to the given topology.

Source reference. Original lemma lemma-constructible-hausdorff-quasi-compact: sources/stacks-topology/pinned-original-source.tex, source lines 4522–4522. Navigation pointer only: the counted text refers to the Stacks item lemma-constructible-hausdorff-quasi-compact, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-fibres-spectral-map-quasi-compact: sources/stacks-topology/pinned-original-source.tex, source lines 4589–4589. Navigation pointer only: the counted text refers to the Stacks item lemma-fibres-spectral-map-quasi-compact, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

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Source reference. Original lemma lemma-image-quasi-compact: sources/stacks-topology/pinned-original-source.tex, source lines 1964–1964. Navigation pointer only: the counted text refers to the Stacks item lemma-image-quasi-compact, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let X and Y be spectral spaces. Let $f : X \to Y$ be a continuous map. Then f is spectral if and only if f is continuous in the constructible topology.*

Proof. The only if part of this is Lemma lemma-fibres-spectral-map-quasi-compact. Assume f is continuous in the constructible topology. Let $V \subset Y$ be quasi-compact open. Then V is open and closed in the constructible topology. Hence $f^{-1}(V)$ is open and closed in the constructible topology. Hence $f^{-1}(V)$ is quasi-compact in the constructible topology as X is quasi-compact in the constructible topology by Lemma lemma-constructible-hausdorff-quasi-compact. Since the identity $f^{-1}(V) \to f^{-1}(V)$ is surjective and continuous from the constructible topology to the usual topology, we conclude that $f^{-1}(V)$ is quasi-compact in the topology of X by Lemma lemma-image-quasi-compact. This finishes the proof. □